Unsupervised Model-Embedded Two-Stage Diffusion Method for Multispectral and Hyperspectral Image Fusion

Jialin Zhou, Shou Feng, Kuo Yuan, Xinlan Xu, Ping Fu, and Jiaqing Qiao

***Abstract*—The multispectral and hyperspectral image (HSI) fusion tasks aim to obtain HSI with a high spatial resolution. However, existing fusion methods usually utilize the degradation simulation for training due to the inaccessible ground truth (GT), which inevitably leads to spatial distortion and parameter bias during application. In addition, neglect of the prior information inherent in inputs and the deep learning (DL) network's low-frequency preference eventually leads to lower interpretability and limited performance. To this end, we proposed a model-embedded two-stage diffusion (MTDiff) method for unsupervised reconstruction of high-spatial resolution HSI. In the first stage, the degradation model is estimated by the prior information inherent input pairs. Meanwhile, in the second stage, embedded with the degradation model, the dual-resolution diffusion model reconstructs a high-resolution HSI (HrHSI). Specifically, treating the degradation process as a fixed diffusion step, an unsupervised paradigm is established through a mapping from upsampled low-resolution HSI to HrHSI. In addition, with the estimated degradation model, the well-designed dual-resolution diffusion model step-by-step perturbs and then denoises the image in both native and degraded resolutions for a fidelity reconstruction with great interpretability. Furthermore, to establish a high-frequency shortcut for network learning, a discrete cosine injection (DCI) module is designed to flatten the frequency information to a clear 2-D domain with a huge high-frequency area, achieving sharp textures and clear structures in fusion results. Extensive systematic experiments across three datasets indicate the superior performance of MTDiff in multispectral and hyperspectral fusion tasks.**

***Index Terms*—Discrete cosine transform (DCT), degradation model, diffusion model, multispectral and hyperspectral image fusion (MHIF).**

I. Introduction

HYPERSPECTRAL image (HSI) captures hundreds of narrow spectral bands, providing a wealth of spectral information about a scene [1]. This technology has revolutionized the acquisition of multiband signatures, making it critical for various domains, including wetland ecological protection [2] and precision agricultural analysis [3]. Nevertheless, there is an inherent tradeoff between the spatial and spectral resolutions imposed by the signal-to-noise ratio constraint of imaging systems [4]. Consequently, HSIs typically suffer from low spatial resolution. In contrast, a high-resolution multispectral image (HrMSI) offers superior spatial resolution at the expense of spectral detail. To overcome the limitations of these individual sensors, multispectral and hyperspectral image fusion (MHIF) [5], [6] has emerged as a pivotal technique to generate a high-resolution HSI (HrHSI). However, the MHIF task is inherently ill-posed due to the significant spatial–spectral differences between the source images [7], [8], which makes high-fidelity reconstruction a formidable challenge [9].

Over the past decades, MHIF methods have evolved from traditional approaches, such as detail-injection [10] and matrix factorization [11], [12] to modern deep learning (DL) techniques [13], [14]. While traditional methods often suffer from spatial artifacts or an over-reliance on handcrafted priors [15], [16], DL-based methods have demonstrated powerful modeling capabilities. However, a dominant class of these methods [17], [18], [19] relies on supervised training with synthetically degraded data. This approach creates a significant domain gap, as the underlying scale-invariance assumption [20] often fails in real-world scenarios, leading to spatial distortion and parameter bias [21], [22]. This has motivated a critical shift toward unsupervised paradigms that can operate at native resolution.

Despite this progress, two fundamental challenges persist in state-of-the-art unsupervised methods. First, many approaches [23], [24] utilize a physical degradation model merely as a terminal constraint within the loss function. This "black-box" treatment fails to deeply integrate physical priors into the generative process, limiting both model interpretability and performance [17]. The network's internal feature transformations remain unguided by physics. Second, deep networks inherently exhibit a low-frequency preference [25], struggling to capture the high-frequency spatial details crucial for high-quality fusion. This inherent bias often results in overly smooth textures and blurry structures, conflicting with the task's objective [26].

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To address these fundamental challenges, we propose a novel model-embedded two-stage diffusion (MTDiff) method for unsupervised MHIF. Our approach is structured as a two-stage pipeline to robustly handle unknown degradations and deeply integrate physical priors. In the first stage, we explicitly estimate the spatial and spectral degradation models directly from the input low-resolution hyperspectral image (LrHSI) and HrMSI pairs. In the second stage, this estimated model is seamlessly embedded into a novel diffusion-based fusion framework. This stage features our core innovations: an unsupervised diffusion paradigm that avoids the need for ground truth (GT), a model-embedded, dual-resolution architecture that provides continuous, per-step guidance during the reverse process, and a discrete cosine injection (DCI) module that explicitly creates a learning shortcut for high-frequency information [27]. This two-stage design not only makes our framework adaptable to real-world scenarios but also enables a deep, physically meaningful integration of prior knowledge into the generative model.

In summary, the contributions of this work are as follows.

1) A novel unsupervised diffusion paradigm is proposed, in which the physical degradation is framed as a fixed, deterministic diffusion step. This formulation directly bridges the distributions of the observable LrHSI and the latent HrHSI, enabling fully unsupervised fusion at native resolution, and thereby circumventing the spatial distortions and parameter biases inherent to supervised methods.
2) A dual-resolution diffusion model is introduced to mathematically integrate the physical degradation model into every step of the diffusion process. Unlike conventional methods that use degradation models merely for terminal loss computation, this per-step integration provides continuous physical guidance, leading to significant improvements in both model interpretability and fidelity.
3) A DCI module is designed to counteract the inherent low-frequency bias of deep neural networks. By establishing a direct shortcut for high-frequency components in the frequency domain, the DCI module enables the method to restore sharp spatial textures and fine-grained structural details.

The remainder of this article is organized as follows. Section II reviews the related works on deep generative methods and DL-based MHIF methods. Section III introduces the proposed MTDiff in detail. Section IV comprehensively presents the experimental results on various datasets. Eventually, Section V summarizes this article.

II. Related Works

This section is devoted to introducing the development of DL-based MHIF methods, following the background and MHIF applications of the diffusion model.

A. DL-Based MHIF Methods

DL has demonstrated its remarkable capacity in nonlinear modeling and feature extraction. With those advantages, it rapidly becomes mainstream in MHIF tasks. Han et al. [28] pioneered the spatial and spectral fusion (SSF)-convolutional neural network (CNN) framework, implementing the DL-based spatial–spectral fusion of multispectral and HSIs. Since that, the DL paradigm has been introduced into MHIF. Subsequently, the approaches include MS-SSFNet by Han et al. [29] and KNLConv by Ran et al. [30], addressing the local receptive field limitations of CNNs via multiscale convolutional kernels and nonlocal convolutional kernels. The advent of transformer architecture has significantly enhanced the long-range modeling capability of deep neural networks. DCTransformer proposed by Ma et al. [31] and pyramid shuffle-and-reshuffle transformer (PSRT) designed by Deng et al. [32] effectively exploits the transformer's long-range modeling capacity and cross-model interactions to achieve superior fusion results. However, these methods mainly treat the deep neural network as a black box for multispectral and hyperspectral modeling, resulting in limited interpretability and limited performance.

Recent unsupervised methods have made significant strides. For instance, Cao et al. [33] designed uHNTC, an unsupervised hybrid network employing model guidance, sufficiently utilizing the capacity of the transformer architecture. Li et al. [34] proposed M2U-Net, achieving unsupervised image fusion via a model-guided deep prior framework. Wang et al. [35] introduced DTDNML, which integrates tensor decomposition and manifold learning for image fusion, realizing unsupervised MHIF via degradation model and reconstruction loss. While these methods enhance interpretability and eliminate the need for GT data, they share a critical limitation: the physical model is only applied to the final output of a one-shot network, typically as a reconstruction loss, which fundamentally treats the physical model as a post hoc validator rather than an active guide during generation. Consequently, the network's internal feature transformations remain a "black-box" unconstrained by physics, limiting the model's ability to capture complex spatial–spectral correlations. In stark contrast, our work introduces a dual-resolution diffusion model where the physical degradation model is mathematically integrated into every step of the diffusion process, providing continuous guidance that yields significant improvements in both fidelity and interpretability.

B. Diffusion Model

The pioneering work of the diffusion model [36], [37] has shifted the paradigm in deep generative methods. As the cutting-edge paradigm in deep generative models, the diffusion model has demonstrated exceptional performance across diverse domains [38]. It aims to establish a mapping from the Gaussian distribution to the target sample distribution, via a fixed forward diffusion process and a DL-based reverse denoising process.

During the forward diffusion process, the distribution of the noisy image at the tth step z_t could be given by the iteration step $t \in (0, T)$ and original image z_0 via the conditional probability $p_\theta(z_{t-1}|z_t, z_0)$. Thus, the noisy image z_t could be mathematically formulated through the following equation:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon \quad \varepsilon \sim N(0, \mathbf{I}) \tag{1}$$

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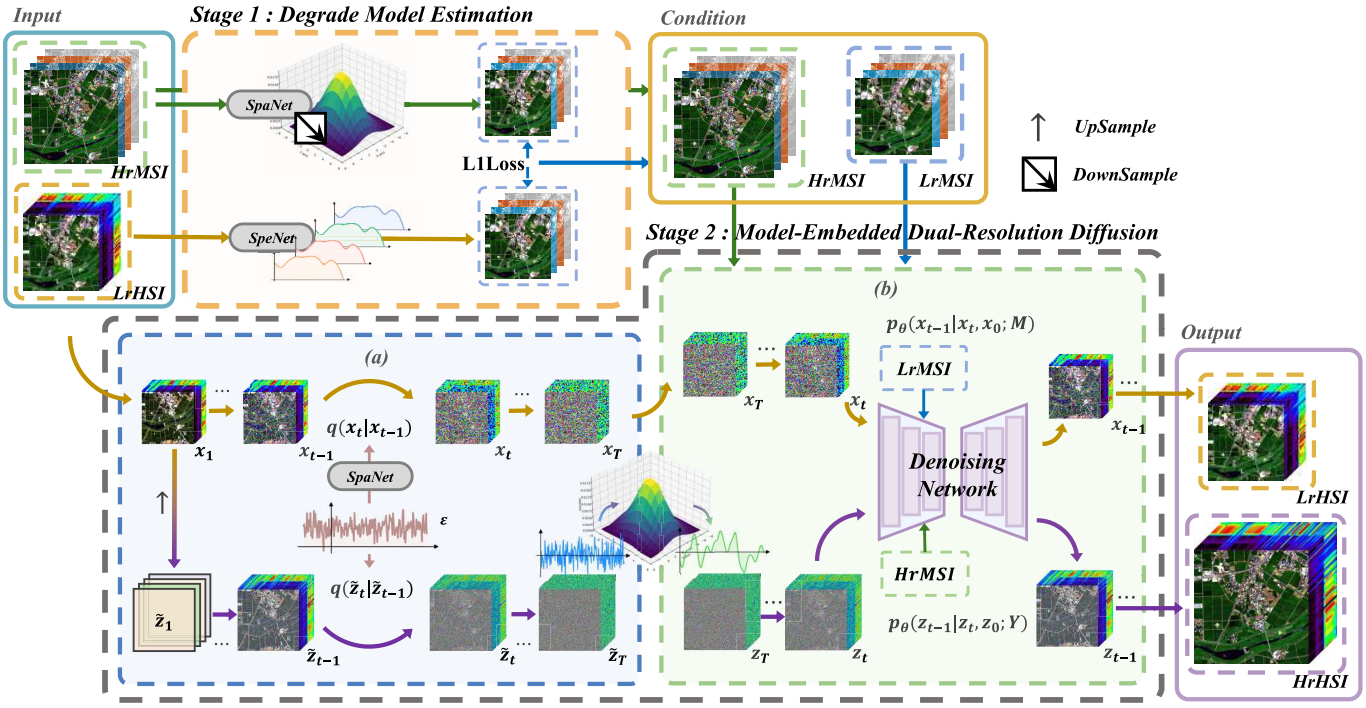


Fig. 1. Overall architecture of the proposed MTDiff framework, featuring two stages. In stage 1, an unsupervised degradation model estimation module learns physical operators by enforcing consistency between degraded modalities. In stage 2, this learned model guides the main pipeline. (a) Dual-resolution forward diffusion process that gradually noises the LrHSI (X) and its upsampled counterpart ($\widetilde{Z}$) and (b) unified reverse denoising process. A shared denoising network, conditioned on HrMSI, and the estimated LrMSI (M), iteratively reconstructs the target HrHSI and LrHSI from pure noise, ensuring high fidelity by deeply integrating these physical priors.

where ε denotes the random noise obeying a standard Gaussian distribution, while α_t serves as the fixed coefficients of the noise scheduler, and $\bar{\alpha}_t$ is defined as $\prod_i^t \alpha_i$.

Meanwhile, in the reverse denoising process, the conditional probability $p_\theta(z_{t-1}|z_t, \widetilde{z}_0)$ can be derived by the Bayes formula. Given the transition from z_t to z_{t-1}, z_0 can be step-by-step generated from the start point z_T. With the conditions of noisy image z_t and origin image $\widetilde{z}_0$ estimated by deep neural network, the noisy image z_{t-1} could be given by

$$z_{t-1} = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \cdot z_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t} \cdot \widetilde{z}_0 + \sqrt{\widetilde{\beta}_t}\varepsilon \quad (2)$$

where $\widetilde{\beta} = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}(1-\alpha_t)$ denotes the variance of the Gaussian distribution, while $\widetilde{z}_0$ is estimated by a deep neural network.

The fixed forward diffusion process establishes a well-defined training track, while the variational optimization enables a theoretical maximum likelihood constraint. Both advantages ensure remarkable training stability compared with GAN. Furthermore, this iterative refinement process is particularly advantageous for high-resolution generation with abundant textural details, making the diffusion model suitable for MHIF tasks. Wu et al. [39] pioneered the integration of the diffusion model into MHIF through their HSR-Diff framework, establishing the first MHIF paradigm via iterative diffusion. Dong et al. [40] developed ISPDiff, which implements multiscale downsampling in the forward diffusion process to capture cross-resolution spatial correlations. Li et al. [41] proposed LDERT, leveraging a latent diffusion paradigm with low-rank tensor decomposition to extract inherent prior knowledge from inputs. Notably, Qu et al. [42] introduced MTLSC-Diff, a multitask architecture that synergizes fusion and classification tasks to guide the reverse denoising process. It ensures the accuracy of downstream tasks. However, the frameworks mentioned above are all supervised, inevitably leading to spatial distortion and parameter bias during application.

III. Proposed Method

The proposed MTDiff method constructs a sophisticated, unsupervised pipeline for MHIF, as illustrated in Fig. 1. Crucially, the entire framework operates without any access to the GT HrHSI, thereby precluding information leakage and ensuring a robust solution. The process begins in Stage 1 with the estimation of the physical degradation model, where SpaNet and SpeNet learn the spatial and spectral operators through a self-referential consistency constraint between the input LrHSI and HrMSI. In Stage 2, these learned operators guide our core innovation: a novel unsupervised diffusion paradigm. We conceptualize the physical degradation as a fixed, deterministic diffusion step, which uniquely allows us to bridge the distributions of the observable and latent data via a single-data distribution shift. This enables fully unsupervised fusion at native resolution. Furthermore, our dual-resolution diffusion model distinguishes itself by mathematically embedding the degradation model into every step of the reverse diffusion process, providing continuous, per-step physical guidance rather than a mere terminal loss constraint.

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The learning is governed by a closed-loop objective: self-supervision at the degraded resolution and a physics-informed loss at the native resolution, which, consistent with the Wald protocol [43], ensures the generated HrHSI degrades back to the original inputs. This section will now detail the problem formulation and each component of this integrated framework.

A. Problem Formulation

The multispectral and HSI fusion task is a highly ill-posed inverse problem, aiming to reconstruct the latent HrHSI $Z \in \mathbb{R}^{H\times W\times C}$ from the input pairs: low-resolution HSI $X \in \mathbb{R}^{h\times w\times C}$ and high-resolution multispectral $Y \in \mathbb{R}^{H\times W\times c}$. Here, H/h, W/w, and C/c represent the spatial dimensions and spectral channels, satisfying the relationships $H \gg h$, $W \gg w$, and $C \gg c$.

The observed LrHSI and HrMSI can be interpreted as spatially and spectrally degraded from the latent HrHSI. Mathematically, this degradation process can be described through a spatial degradation governed by the point spread function (PSF) and a spectral degradation operator governed by the spectral response function (SRF). The model, termed the degradation model, is formulated as follows:

$$\begin{cases} X = DZ + N_P \\ Y = ZR + N_S \end{cases} \tag{3}$$

where X denotes LrHSI, Y represents HrMSI, and Z serves as the latent HrHSI. The spatial degradation operator D contains a blurring kernel and a downsampling operator governed by PSF, and the spectral degradation operator R is characterized by a spectral transition matrix governed by SRF. N_P and N_S represent the noise introduced during spatial and spectral degradation processes.

The degradation model concisely establishes the relationship between the latent HrHSI and observed LrHSI or HrMSI via PSF or SRF. Based on the degradation model in (3), the reconstruction constraint is formulated as follows:

$$\min \|X - DZ\|_2 + \|Y - ZR\|_2 + \beta\mathcal{R}(Z) \tag{4}$$

where $\|\cdot\|_2$ denotes as $L2$-norm, β serves as a nonnegative regularization coefficient, and $\mathcal{R}(Z)$ represents other regularzation terms.

B. Degradation Model Estimation

To extract the physical constraints from the input pair, we first estimate the degradation model, as formalized in (3). A critical challenge in such estimation is avoiding arbitrary, non-physical solutions that overfit to the test data. Inspired by the principle of model parameterization for degradation estimation [44], which suggests describing complex blur phenomena through simple, parameterized models (e.g., a Gaussian kernel defined by its standard deviation), we adopt a similar philosophy. Instead of learning unconstrained operators, we design SpaNet and SpeNet with strong architectural priors that mimic established forward models in remote sensing.

SpaNet approximates the spatial degradation using a single convolutional layer. Crucially, its kernel is constrained by nonnegativity and energy conservation, effectively parameterizing it as a valid PSF rather than an arbitrary filter. Similarly, SpeNet models the spectral degradation as a linear transformation via a 1×1 convolutional layer with a nonnegativity constraint, reflecting the physical process of spectral integration. This principled approach of embedding physical knowledge into deliberately simple network architectures ensures the estimated degradation model is generalizable, interpretable, and physically meaningful. We then impose these learned operators on the input pairs LrHSI and HrMSI, formulated as follows:

$$\begin{cases} M_1 = X \cdot \widetilde{R} = DZ \cdot \widetilde{R} \\ M_2 = \widetilde{D} \cdot Y = \widetilde{D} \cdot ZR \end{cases}$$
$$\widetilde{D} := \varphi_{\theta_{\text{SpaNet}}}$$
$$\widetilde{R} := \varphi_{\theta_{\text{SpeNet}}} \tag{5}$$

where M_1 denotes LrMSI obtained by spectral degrading from the LrHSI X, while M_2 represents LrMSI generated by spatial degradation from Y. The operators $\widetilde{D}$ and $\widetilde{R}$ are the approximations learned by the CNNs SpaNet and SpeNet. Meanwhile, $\varphi_{\theta_{\text{SpaNet}}}$ and $\varphi_{\theta_{\text{SpeNet}}}$ denote the learnable parameters of SpaNet and SpeNet.

Theoretically, two degraded images, M_1 and M_2, should have the same radiometric and spectral properties. Thus, this consistency enables an unsupervised constraint for SpaNet and SpeNet learning, formulated as follows:

$$\min \left|\left|X \cdot \widetilde{R} - \widetilde{D} \cdot Y\right|\right|_1. \tag{6}$$

The estimated degradation model represented by SpaNet and SpeNet could extract the shared redundant information LrMSI from the input pairs HrMSI and LrHSI, formulated as follows:

$$\begin{aligned} M &= \frac{1}{2}M_1 + \frac{1}{2}M_2 \\ &= \frac{1}{2}\varphi_{\theta_{\text{SpeNet}}}(X) + \frac{1}{2}\varphi_{\theta_{\text{SpaNet}}}(Y) \end{aligned} \tag{7}$$

where M denotes the redundant information LrMSI.

C. Dual-Resolution Diffusion Model

While diffusion models excel at high-fidelity image synthesis, the standard formulation is inherently supervised, requiring paired clean and noisy data for training. This presents a fundamental conflict in the context of MHIF, where the GT HrHSI, denoted by z_0, is by definition unavailable.

To bridge this gap, we introduce MTDiff, a novel framework that refines the diffusion process for fully unsupervised learning by explicitly embedding the physical degradation model into it. Our central insight is to reframe the known spatial degradation as a deterministic diffusion step. This reframe establishes a critical link between the data we possess (LrHSI) and the data we seek (HrHSI). Specifically, it allows us to initiate a forward diffusion process in the native-resolution space starting from the upsampled LrHSI instead of the unknown z_0 itself. The carefully formulated data distribution shift aligns the terminal noise distribution of the upsampled LrHSI with the distribution that would have resulted from

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the latent z_0. Guided by physical consistency constraints, this alignment creates a valid starting point for a reverse denoising process that reconstructs z_0.

To formalize this unsupervised strategy, we first begin by defining the standard forward diffusion process for the latent HrHSI z_0 in a hypothetical supervised setting. Derived from (1), this process is formulated as follows:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon \tag{8}$$

where z_0 denotes the latent HrHSI and z_t represents the noisy image at the tth step.

1) Forward Diffusion Progress: The forward diffusion process is first formalized in the degraded-resolution domain. By applying the spatial degradation operator D to the hypothetical native-resolution process in (8), we can describe the noising process for the observable LrHSI, x_0

$$\begin{aligned} x_t &= D \cdot z_t \\ &= \sqrt{\bar{\alpha}_t} D \cdot z_0 + \sqrt{1-\bar{\alpha}_t} D \cdot \varepsilon \\ &= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t} D \cdot \varepsilon \\ \Rightarrow x_t &= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t} D \cdot \varepsilon \end{aligned} \tag{9}$$

where D is the spatial-degraded operator approximation of the estimated degradation model, while x_0 represents the LrHSI and x_t serves as the noisy LrHSI at the tth step.

Next, the parallel process is constructed in the native-resolution domain. As the true HrHSI z_0 is inaccessible, we use its physically related and observable proxy: the upsampled LrHSI, denoted by $x_0 \uparrow$. This proxy is related to z_0 via the blurring operator B, such that $x_0 \uparrow = B \cdot z_0$

$$\begin{aligned} \widetilde{z}_1 &= B \cdot z_0 = x_0 \uparrow \\ x_1 &= I \cdot x_0. \end{aligned} \tag{10}$$

With these accessible starting points, $\widetilde{z}_1$ and x_1, we can now define the complete dual-resolution forward diffusion process for $t \in [1, T]$

$$\begin{aligned} \widetilde{z}_t &= \sqrt{\bar{\alpha}_t}\widetilde{z}_1 + \sqrt{1-\bar{\alpha}_t} \cdot \varepsilon \\ x_t &= \sqrt{\bar{\alpha}_t} x_1 + \sqrt{1-\bar{\alpha}_t} D \cdot \varepsilon \end{aligned} \tag{11}$$

where $\widetilde{z}_t$ represents the noisy state in our native-resolution proxy process. These two coupled equations form the foundation of our unsupervised learning framework.

2) Reverse Denoising Process: The forward processes define how to add noise, while the reverse process learns to remove it to reconstruct z_0. The key to enabling this in an unsupervised manner lies in formalizing the data distribution shift. This involves establishing a mathematical link between the terminal state of our accessible proxy process, $\widetilde{z}_T$, and the terminal state of the latent process, z_{T-1}.

To create this link, we add one more step of the final latent state z_{T-1} by incorporating the blurring operator B. This data distribution shift is crucial for alignment

$$\begin{aligned} z_T &= B \cdot z_{T-1} \\ &= \sqrt{\bar{\alpha}_t} B \cdot z_0 + \sqrt{1-\bar{\alpha}_t} B \cdot \varepsilon. \end{aligned} \tag{12}$$

With this, we can derive a direct relationship between z_T and $\widetilde{z}_T$. By combining (10)–(12), we find

$$\begin{aligned} z_T &= \sqrt{\bar{\alpha}_t} B \cdot z_0 + \sqrt{1-\bar{\alpha}_t} B \cdot \varepsilon \\ &= \sqrt{\bar{\alpha}_t}\widetilde{z}_1 + \sqrt{1-\bar{\alpha}_t} B \cdot \varepsilon \\ &= \widetilde{z}_T + \sqrt{1-\bar{\alpha}_t}\,(B - I) \cdot \varepsilon. \end{aligned} \tag{13}$$

This equation mathematically shows the data distribution shift. It reveals that the target distribution of z_T can be reached from the accessible distribution of $\widetilde{z}_T$ via a simple, noise-dependent correction term.

The final reverse process is shown in (14), which is enabled by the shift. This alignment provides a valid starting point for the dual-resolution reverse denoising process. We can now iteratively denoise from a random sample $\widetilde{z}_T \sim \mathcal{N}(0, I)$ back to step $t = 0$. The reverse step is formulated as follows:

$$\begin{aligned} z_{t-1} &= \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \cdot z_t + \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\,(1-\alpha_t) \cdot \hat{z}_0 + \sqrt{\widetilde{\beta}_t}\varepsilon \\ x_{t-1} &= \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \cdot x_t + \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\,(1-\alpha_t) \cdot \hat{x}_0 \\ &\quad + \sqrt{\widetilde{\beta}_t} D \cdot \varepsilon \end{aligned} \tag{14}$$

where the estimations for the clean images, $\hat{z}_0$ and $\hat{x}_0$, are predicted by a shared denoising network $\mathcal{F}_\theta$ conditioned on the available multispectral data

$$\begin{cases} \hat{z}_0 = \mathcal{F}_\theta\,(z_t, t; X \uparrow, Y) \\ \hat{x}_0 = \mathcal{F}_\theta\,(x_t, t; X, M) \end{cases} \tag{15}$$

where $\hat{z}_0$ and $\hat{x}_0$ denote the result generated by denoising network from z_t and x_t, X represents LrHSI, and Y denotes HrMSI. While M serves as the LrMSI obtained by the estimated degradation model.

D. Denoising Network

In the reverse denoising process of MTDiff, the accurate noise estimation is obtained through the deep neural network, which enables the reconstruction of HrHSI.

1) U-Net Architecture: Compared with the image features, the Gaussian noise is high-frequency information, requiring specialized architectural considerations. Unlike the residual structures, the U-Net architecture with skip connections demonstrates superior capacity for high-frequency information modeling.

As illustrated in Fig. 2, the denoising network adopts a hierarchical U-Net structure inspired by BigGAN [45]. The encoder is constructed by U-Net denoising blocks (UDBs) with spatial downsampling operators, while the decoder is formed by UDBs and upsampling operators.

During the encoding process, the feature compression is realized by spatial downsampling and channelwise expansion. Notably, the self-attention mechanism is imposed at the end of the encoder to construct long-range nonlocal features, enhancing the global context information. As for the decoder, the image is reconstructed through the UDBs and upsampling operators served by transposed convolutions. Due to the skip

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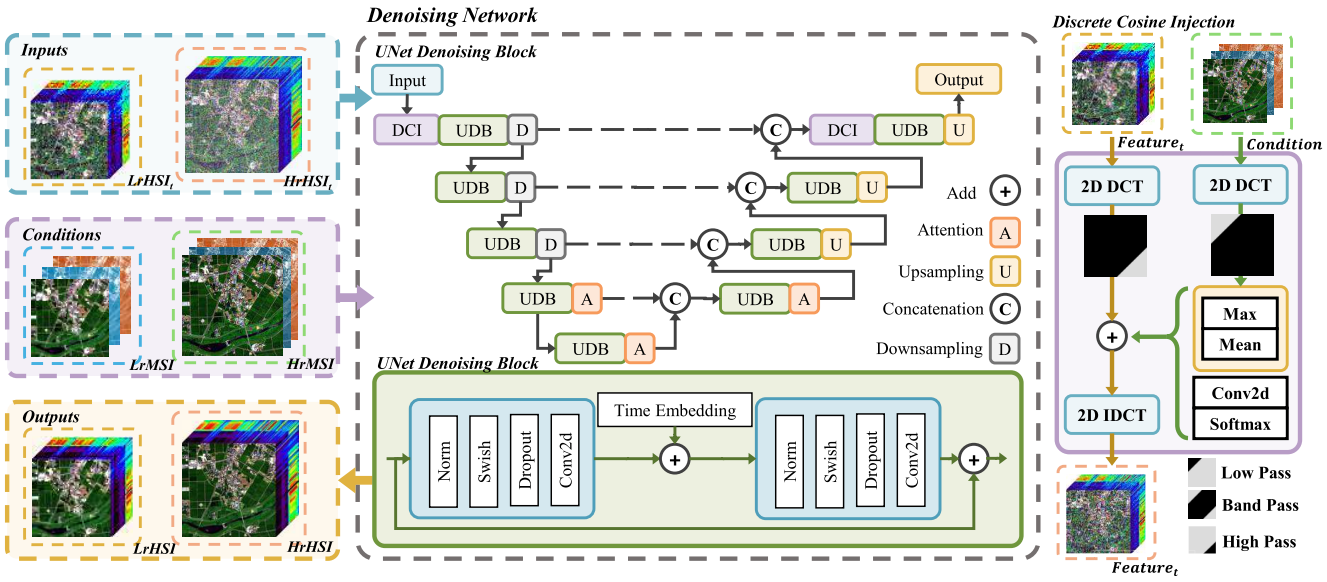


Fig. 2. Detailed architecture of our U-Net-based denoising network, which processes multiscale features via skip connections. A key innovation is the DCI module, which explicitly creates a high-frequency shortcut to counteract the network's inherent low-frequency preference. By operating in the 2-D discrete cosine transform (DCT) domain, the DCI module selectively injects sharp structural details from the HrMSI condition into the generation process, which is critical for reconstructing fine textures and sharp edges in the final fused image.

connections from the encoder directly to the decoder, high-frequency information is transmitted effectively, ensuring the modeling capacity for high-frequency Gaussian noise.

As the timestamp t steps forward, the noise variance generated by the noise scheduler α_t increases. Thus, an effective method is needed to embed the timestamp t into UDBs. To address this requirement, the cosine positional embedding is adopted, which is formulated as follows:

$$\begin{cases} \mathcal{PE}(t)_{2i} = \sin \dfrac{t}{10000^{2i/d}} \\ \mathcal{PE}(t)_{2i+1} = \cos \dfrac{t}{10000^{2i+1/d}} \end{cases} \tag{16}$$

where $\mathcal{PE}$ denotes the positional embedding feature vectors, d represents the dimension of embedding vectors, and i serves as the ith dimension of the feature vectors. Leveraging the cosine positional embedding, the denoising network could model the Gaussian noise in the tth step more accurately, improving the fidelity of the fusion outputs.

2) Discrete Cosine Injection: MTDiff achieves unsupervised image fusion via embedding a degradation model into a diffusion framework. However, the DL networks prefer low-frequency information while objective fitting, leading to a conflict between the DL networks' low-frequency preference and the requirement for high-frequency details of HrHSI reconstruction. To address it, the DCI module shown in Fig. 2 is introduced to establish a high-frequency shortcut for learning, achieving sharp textures and clear structures in the fusion result.

For the noisy images x_t and z_t at the timestamp t, the DCI module first imposed a 2-D discrete cosine transform on them to decompose the spatial frequency information. Subsequently, binary masking filters are applied to separate different frequency parts. The binary maskers are specifically designed, formulated as follows:

$$\begin{cases} \mathcal{M}_{\text{lp}}(x,y) = 1, & \text{if } (x+y) \le f_{\text{lp}}, \quad \text{else } 0 \\ \mathcal{M}_{\text{bp}}(x,y) = 1, & \text{if } f_{\text{lp}} < (x+y) \le f_{\text{hp}}, \quad \text{else } 0 \\ \mathcal{M}_{\text{hp}}(x,y) = 1, & \text{if } f_{\text{hp}} < (x+y), \quad \text{else } 0 \end{cases} \tag{17}$$

where f_{lp} and f_{hp} denote the low and high cutoff frequencies of the 2-D discrete transform domain. Leveraging these binary filters $\mathcal{M}_{\text{lp}}$, $\mathcal{M}_{\text{bp}}$, $\mathcal{M}_{\text{hp}}\}$, the DCI module realizes the extraction of low-frequency feature in tth step noisy HSI and high-frequency spatial texture feature in condition MSI. Given the high-frequency information scattering in the 2-D discrete transform domain, spatial self-attention is employed for the construction of long-range correlations. After high-frequency features are injected into the low-frequency domain, a 2-D inverse discrete transform is employed to generate the tth step noisy image with high-frequency feature injection.

E. Loss Function

MTDiff achieves unsupervised fusion via a two-stage pipeline. First, the estimation of the degradation model is performed based on the input pairs LrHSI and HrMSI. Then, a dual-resolution diffusion model is constructed with it embedded, generating the latent HrHSI. As for the estimation of the degradation model, based on (6), the loss function could be formulated as follows:

$$\mathcal{L}_1 = \left\| \varphi_{\theta_{\text{SpeNet}}}(X) - \varphi_{\theta_{\text{SpaNet}}}(Y) \right\|_1. \tag{18}$$

For the dual-resolution diffusion model, MTDiff conducts learning in both native resolution and degraded resolution. Thus, the loss function consists of the native-resolution loss and the degraded-resolution loss, formulated as follows:

$$\mathcal{L}_2 = \alpha \mathcal{L}_{\text{degraded}} + \beta \mathcal{L}_{\text{origin}} \tag{19}$$

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TABLE I
PARAMETERS AND SIMULATION RESULTS OF THREE BENCHMARK DATASETS

Datasets	Chikusei	Houston	Washington DC
Bands	128	48	191
Spatial Resolution(m)	2.5	1	—
Spectral Range(nm)	363∼1018	380∼1050	400∼2400
Size of HrHSI	896×256	896×128	896×256

where $\mathcal{L}_{\text{degraded}}$ denotes the degraded-resolution loss, and $\mathcal{L}_{\text{origin}}$ represents the native-resolution loss. The weight coefficients α and β of them satisfy $\alpha + \beta = 1$.

At the degraded resolution, the denoising learning process could be guided directly by the GT LrHSI, as a supervised learning paradigm. Derived from the energy constraint formula obtained via the KL divergence, the degraded-resolution loss function is formulated as follows:

$$\mathcal{L}_{\text{degraded}} = \|X - \mathcal{F}_\theta(x_t, t; X, M)\|_2 . \tag{20}$$

However, at the native resolution, due to the unavailable of latent HrHSI, the denoising learning process integrates the degradation model as the constraint formula. The loss function at the native resolution is formulated as follows:

$$\begin{aligned}\mathcal{L}_{\text{origin}} = &\widetilde{\alpha}\left\|X - \varphi_{\theta_{\text{SpaNet}}}(\mathcal{F}_\theta(\widetilde{z}_t, t; X\uparrow, Y))\right\|_2 \\ &+ \widetilde{\beta}\left\|Y - \varphi_{\theta_{\text{SpeNet}}}(\mathcal{F}_\theta(\widetilde{z}_t, t; X\uparrow, Y))\right\|_2\end{aligned} \tag{21}$$

where $\widetilde{\alpha}$ and $\widetilde{\beta}$ denote the weight of spectral degradation constraint and spatial degradation constraint, which satisfy $\widetilde{\alpha} + \widetilde{\beta} = 1$.

IV. EXPERIMENT

In this section, extensive experiments have been conducted, including performance comparison, ablation studies, and robustness studies, to validate the effectiveness of the proposed method.

A. Datasets and Experimental Settings

1) Datasets: In this study, three benchmark datasets are employed to evaluate the performance of all methods, including the Chikusei [46], Houston [47], and Washington DC Mall. Specifically, the Chikusei dataset is captured by the Headwall Hyperspec-VNIR-C, providing 128 spectral bands from 363 to 1018 nm and a 2.5-m ground sampling distance. The Houston dataset acquired by the ITRES CASI 1500 has 48 spectral bands from 380 to 1050 nm and a 1-m ground sampling distance. The Washington DC Mall dataset is taken by the HYDICE sensor, providing 191 spectral channels available. These datasets are captured from various sensors in different spectral band ranges, ground sampling distances, and spatial resolutions, which allow us to sufficiently evaluate the proposed method in different data distributions.

Due to the unavailable of HrHSI in practice, a simulation operation is imposed on those datasets to generate the input pairs LrHSI and HrMSI. The LrHSI is acquired through a Gaussian blurring kernel followed by a downsampling operation, and the HrMSI is generated by the practical SRF of QuickBirds. The parameters of the datasets and the shape of simulation results are shown in Table I

2) Evaluation Metrics: Six quantitative metrics are employed to quantitatively evaluate the fusion result in spatial and spectral qualities. Specifically, the universal quality index (UQI) [48] and the structural similarity (SSIM) [49] evaluate the spatial quality of the fusion outcomes. The ideal value of them is both one. Meanwhile, the peak signal-to-noise ratio (PSNR) [49], the spectral angle mapper (SAM) [50], and the root-mean-square error (RMSE) give the spectral error in the fusion result. The ideal value of PSNR is infinity. As for SAM and RMSE, the ideal values are 0. What is more, the Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) [51] assesses the fusion outputs in both spatial and spectral aspects, whose ideal value is 0. For the visual evaluation, an SAM heatmap, a mean relative absolute error (MRAE) heatmap, and a residual heatmap are employed to exhibit the distortion in fusion results.

3) Implementing Details: Our proposed MTDiff and compared methods are both running on an Intel Core i9-12900K CPU with a GeForce GTX 3090 Ti. The optimization of MTDiff is conducted in PyTorch 2.5.1 under the Python 3.11 framework. MTDiff is optimized in two stages. During the first stage, the degradation model estimation is conducted via a learning rate of 0.0001 for 30 000 iterations with the AdamW optimizer. In the second stage, embedded with the estimated degradation model, the dual-resolution diffusion model employs 1600 steps for the forward diffusion process and sampling. The AdamW optimizer with a learning rate of 0.0001 for 20 000 iterations is adopted for training. As for the loss functions, the weights in (19) are both set to 0.5, balancing the guidance from both resolutions. Similarly, in (21), the weights are also set to 0.5, giving equal consideration to spatial and spectral fidelity. For a fair comparison, we follow the original settings in the official implementations of each compared method.

B. Quantitative Performance Assessments

We select 11 various methods, from unsupervised to supervised, as competitors to assess the superior performance of our proposed method. Both conventional and DL based are adopted to evaluate the performance of our proposed method. Conventional methods consist of SFIMHS [52], GLPHS [53], CNMF [12], FUSE [54], and HySure [55]. Meanwhile, from supervised to unsupervised, the DL-based methods include four unsupervised methods UDALN [56], SURE [57], M2U-Net [34], CS2DIPs [58], and two supervised method PSRT [32] with a diffusion method S^2CycleDiff [59], which could sufficiently reveal the outstanding performance of our proposed method.

1) Results on the Chikusei Dataset: Our evaluation commenced on the Chikusei dataset, utilizing an 896 × 256 image patch from coordinate (512, 512), which is characterized by a complex mixture of dense residential areas and varied agricultural plots. The visual results are presented in Fig. 3. As observed, many methods struggle with this scene's heterogeneity; for instance, conventional methods like SFIMHS produce significant spatial and spectral distortions, while even some DL

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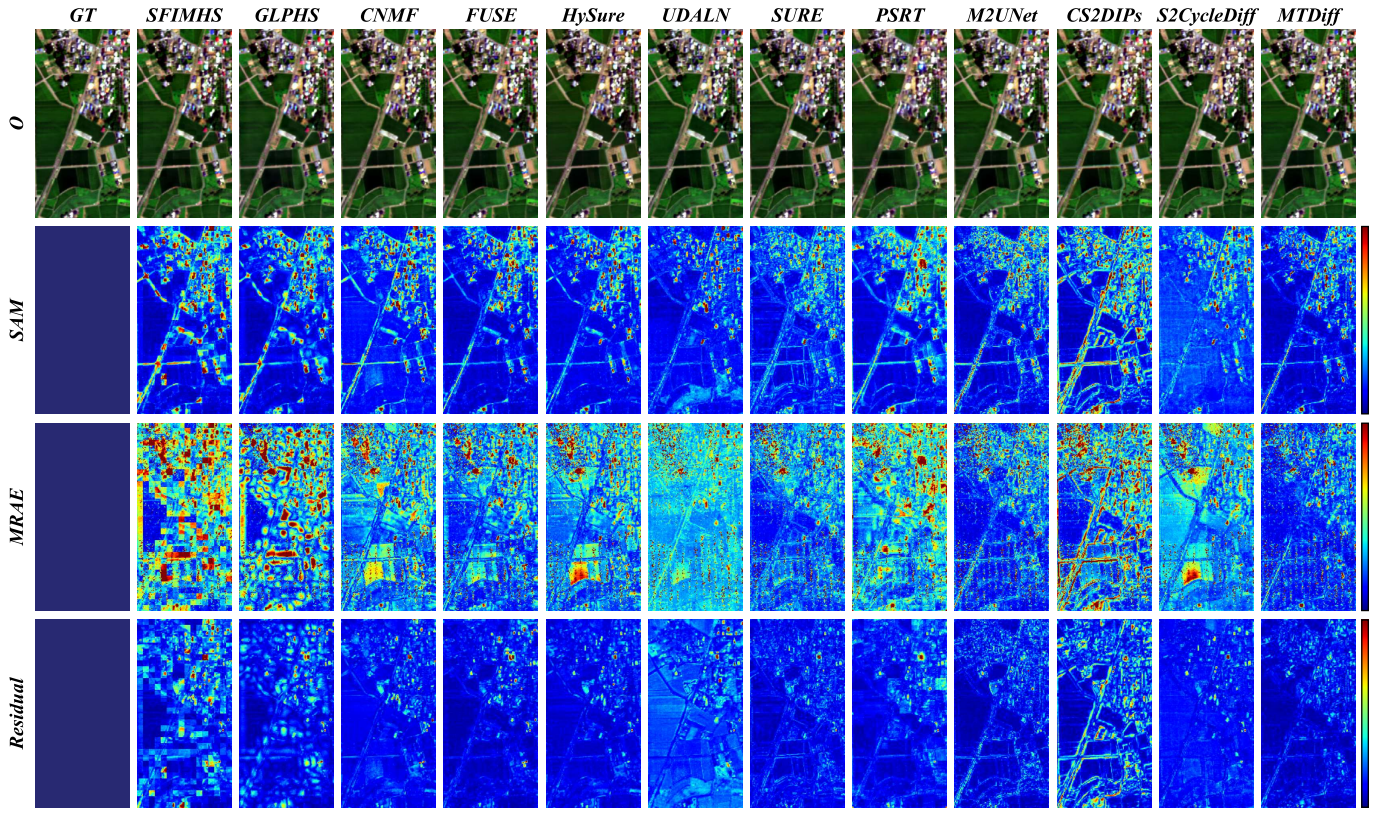


Fig. 3. Visual comparison of all fusion methods on the Chikusei dataset. The top row (O) displays the pseudo-color GT image and the fusion results from 12 different methods. The subsequent rows show the corresponding error heatmaps for SAM, MRAE, and pixelwise residual error.

TABLE II

SIX QUANTITATIVE METRICS ON THE CHIKUSEI DATASET. THE BEST VALUES ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST VALUES ARE HIGHLIGHTED IN BLUE

Method	PSNR	RMSE	SSIM	SAM	UQI	ERGAS
BEST	∞	0	1	0	1	0
SFIMHS	17.9149	0.1271	0.9542	4.3269	0.9461	4.2970
GLPHS	35.2825	0.0172	0.9737	2.5432	0.9850	1.3455
CNMF	36.3758	0.0152	0.9765	2.8174	0.9838	1.3631
FUSE	37.6098	0.0132	0.9760	2.6239	0.9863	1.3371
HySure	37.2191	0.0138	0.9583	3.3732	0.9855	1.3095
UDALN	33.1052	0.0221	0.9837	2.1488	0.9837	1.4108
SURE	37.5075	0.0133	0.9762	2.2618	0.9858	1.2335
PSRT	35.5014	0.0168	0.9661	2.9752	0.9763	1.8172
M2U-Net	37.0365	0.0141	0.9758	2.1113	0.9872	1.1065
CS2DIP	31.6617	0.0261	0.9504	3.1006	0.9720	1.7019
S^2CycleDiff	34.5959	0.0186	0.9698	3.8637	0.9790	1.4522
MTDiff	38.8624	0.0114	0.9821	1.9863	0.9894	1.0130

models exhibit blurring artifacts at the edges of structures. In contrast, our MTDiff method successfully reconstructs sharp details and accurate spectral signatures, yielding a visually superior result. This is further validated by the error heatmaps, where MTDiff exhibits minimal error, particularly preserving crisp edges along roads and buildings.

The quantitative analysis in Table II confirms MTDiff's superior performance. Our method achieves the best values on five of the six metrics (PSNR, RMSE, SAM, UQI, and ERGAS), demonstrating a remarkable balance of spatial and spectral fidelity. However, it is noteworthy that UDALN achieves the highest SSIM score. We attribute this to different inductive biases of the underlying models. SSIM is highly sensitive to local SSIM. Our diffusion-based MTDiff excels at modeling the global data distribution, leading to reconstructions with the superior overall fidelity and spectral accuracy. This focus on the global consistency might result in subtle, naturalistic smoothing of highly stochastic local textures, leading to a slightly lower SSIM score. This highlights a fundamental tradeoff between local texture replication and global data distribution fidelity.

2) Results on the Houston Dataset: The fusion experiments were extended to the Houston dataset, using a 896 $\times$ 128 urban image patch from coordinate (0, 0), which features a variety of building materials and vegetation. The visual results are presented in Fig. 4. As observed, many DL methods produce noticeable artifacts; for instance, PSRT introduces color distortion on building roofs, and UDALN's result appears overly smooth, sacrificing fine textural details on the ground and rooftops. In contrast, our MTDiff method successfully reconstructs sharp structural details and accurate colors, yielding a visually superior result. This is further validated by the error heatmaps, where MTDiff exhibits the lowest error in both the MRAE and residual maps, indicating exceptional pixel-level accuracy.

The quantitative analysis in Table III confirms MTDiff's superior performance. Our method achieves the best values on five of the six metrics (PSNR, RMSE, SSIM, UQI, and ERGAS), demonstrating a remarkable balance of spatial and

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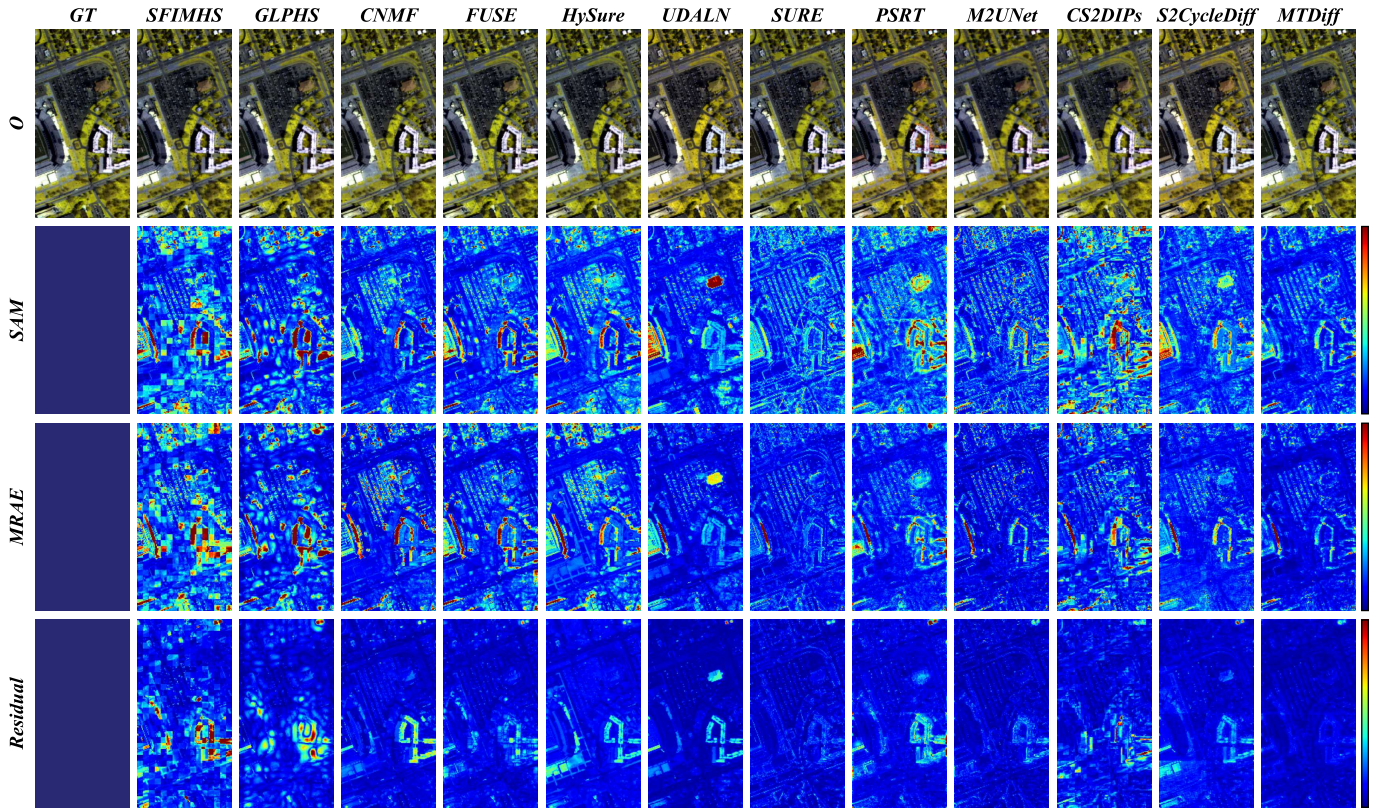


Fig. 4. Visual comparison of all fusion methods on the Houston dataset. The top row (O) displays the pseudo-color GT image and the fusion results from 12 different methods. The subsequent rows show the corresponding error heatmaps for SAM, MRAE, and pixelwise residual error.

TABLE III
SIX QUANTITATIVE METRICS ON THE HOUSTON DATASET. THE BEST VALUES ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST VALUES ARE HIGHLIGHTED IN BLUE

Method	PSNR	RMSE	SSIM	SAM	UQI	ERGAS
BEST	∞	0	1	0	1	0
SFIMHS	31.3432	0.0271	0.9713	3.0223	0.9739	1.8019
GLPHS	35.1055	0.0176	0.9839	2.8085	0.9834	1.2904
CNMF	37.4400	0.0134	0.9864	2.2028	0.9848	1.2408
FUSE	38.2283	0.0123	0.9855	2.5005	0.9835	1.6741
HySure	36.8686	0.0143	0.9859	2.7567	0.9826	1.3007
UDALN	40.5327	0.0094	0.9900	1.9638	0.9792	1.9086
SURE	40.4012	0.0095	0.9823	2.2985	0.9833	1.2550
PSRT	35.5262	0.0167	0.9821	2.8846	0.9767	1.6936
M2U-Net	40.3491	0.0096	0.9847	2.3352	0.9870	0.9846
CS2DIP	36.1724	0.0155	0.9794	3.2926	0.9771	1.4363
S^2CycleDiff	38.2219	0.0123	0.9887	2.3338	0.9831	1.1982
MTDiff	43.0591	0.0070	0.9915	1.9874	0.9892	0.9388

spectral fidelity. UDALN obtains the best SAM score in this experiment. The SAM exclusively measures the angular difference between spectral vectors, ignoring their magnitude. Our MTDiff aims to minimize a loss function that balances both spectral and spatial errors in a Euclidean sense, leading to excellent radiometric performance and overall quality, but with a marginal tradeoff in the spectral angle for some pixels. This suggests that for applications where preserving the exact spectral angle is the single most critical factor, and other specialized models might be competitive.

3) Results on the Washington DC Mall Dataset: To assess model robustness, we performed experiments on the challenging Washington DC Mall dataset, using an 896 $\times$ 256 image patch renowned for its large, uniform structures and subtle spectral variations, from coordinate (256, 16). The visual results are presented in Fig. 5. The performance disparity is striking: many established algorithms, including both conventional (FUSE, HySure) and DL-based (UDALN, S^2CycleDiff), suffer from severe spectral distortion and spatial artifacts, indicating a lack of robustness. In contrast, our MTDiff method produces a result of exceptional quality, with precise color rendition and sharp details that are visually superior to all competitors. This is powerfully demonstrated in the error heatmaps, where MTDiff shows minimal deviation from the GT.

The quantitative analysis in Table IV provides decisive evidence of this superiority. Our method achieves a remarkable achievement, ranking first across all six metrics, often by a substantial margin. This comprehensive success on a dataset where other methods falter strongly validates the architectural robustness of MTDiff. The deep, per-step integration of the physical degradation model acts as a powerful regularizer, preventing the model from diverging in challenging scenarios and consistently guiding it toward a high-fidelity solution. In this case, MTDiff does not present a tradeoff but rather a comprehensive improvement, establishing a new state-of-the-art benchmark for this dataset.

4) Analysis of Spectral Curves: In addition to spatial fidelity, preserving spectral consistency is a critical benchmark for fusion quality. As illustrated in Fig. 6, where the

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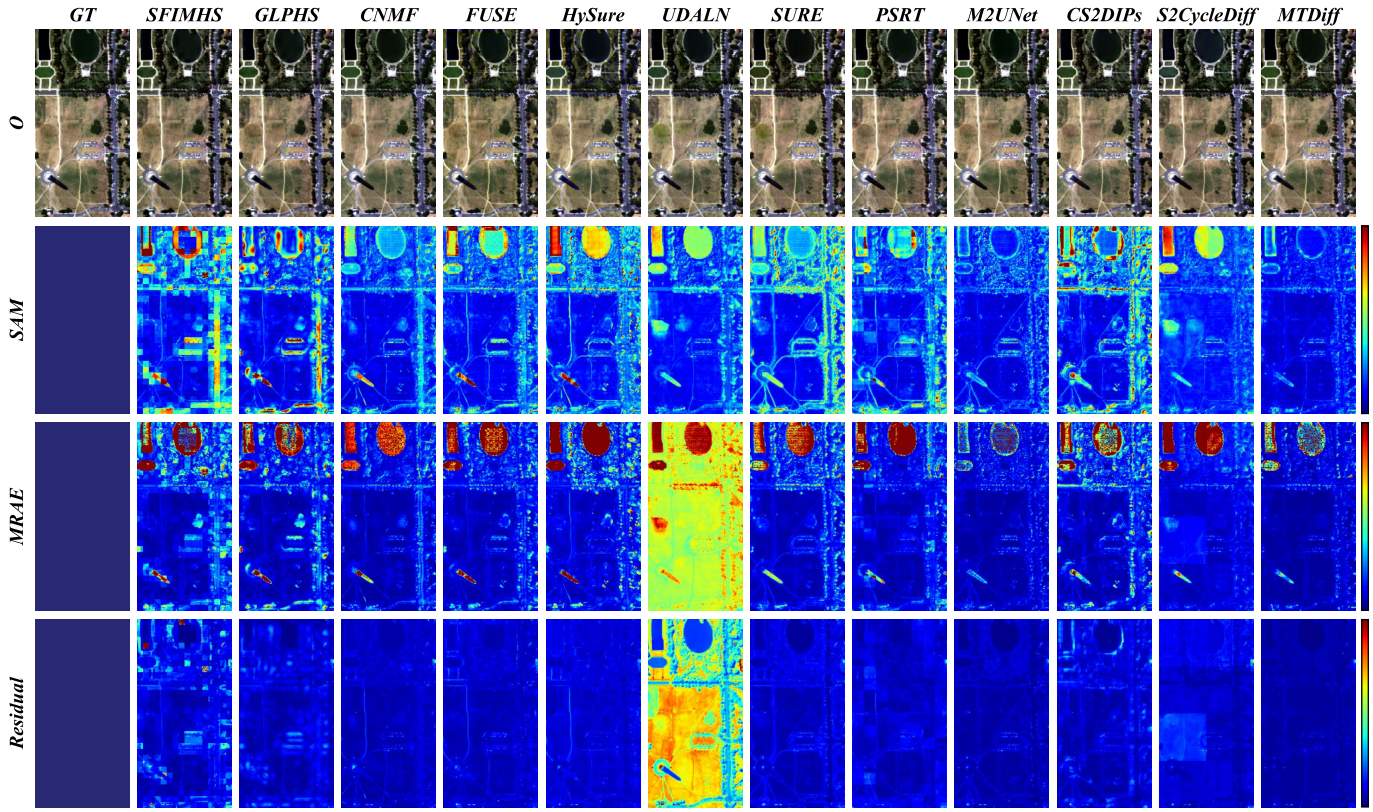


Fig. 5. Visual comparison of all fusion methods on the Washington DC Mall dataset. The top row (O) displays the pseudo-color GT image and the fusion results from 12 different methods. The subsequent rows show the corresponding error heatmaps for SAM, MRAE, and pixelwise residual error.

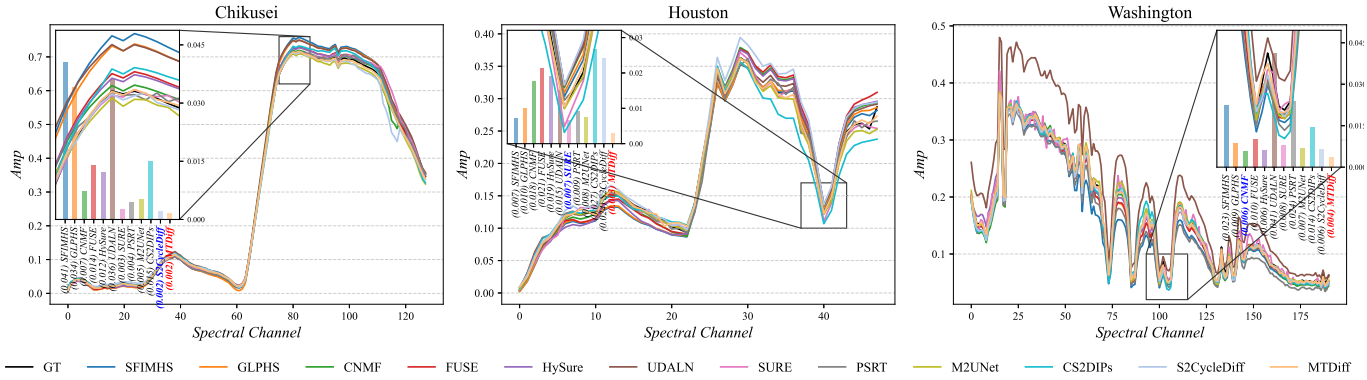


Fig. 6. Qualitative and quantitative comparisons of spectral fidelity for a selected pixel from each benchmark dataset. The main plots show the reconstructed spectral reflectance curves from all methods against the GT. To provide a quantitative error analysis, the key turning points are magnified. The bar charts represent the absolute spectral error for each method at a specific, representative spectral band, offering a direct numerical comparison of reconstruction accuracy. The best results are highlighted in red. The second-best results are highlighted in blue. The selected pixel coordinates (row, column) are (83, 158) for Chikusei, (109, 37) for Houston, and (206, 137) for Washington DC Mall.

spectral curves of a selected pixel from each dataset are plotted, our proposed MTDiff demonstrates superior spectral fidelity. While most methods capture the general spectral trend, they often struggle with significant deviations at critical turning points and absorption features. In contrast, MTDiff's reconstruction most accurately traces these complex variations across all datasets.

To substantiate this qualitative observation with quantitative evidence, the magnified insets display bar charts representing the absolute spectral error for each method at a representative band. These charts unequivocally show that MTDiff achieves one of the lowest error magnitudes, confirming its minimal spectral distortion. This precise approximation of spectral features is paramount for downstream applications, such as material classification and change detection, as it directly enhances the reliability and effectiveness of the fused data.

C. Ablation Study

This sections aim to present the result of an ablation study, which assesses the performance of the dual-resolution diffusion paradigm, and the DCI module in the MTDiff. The

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TABLE IV
SIX QUANTITATIVE METRICS ON THE WASHINGTON DC MALL DATASET. THE BEST VALUES ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST VALUES ARE HIGHLIGHTED IN BLUE

Method	PSNR	RMSE	SSIM	SAM	UQI	ERGAS
BEST	∞	0	1	0	1	0
SFIMHS	12.8888	0.2268	0.9612	3.7376	0.9678	18.3414
GLPHS	36.3289	0.0153	0.9791	2.6726	0.9891	1.2523
CNMF	38.8928	0.0114	0.9814	2.3020	0.9917	1.1049
SURE	37.5023	0.0133	0.9718	2.8455	0.9891	1.2611
HySure	37.3604	0.0136	0.9450	3.2582	0.9874	1.3922
UDALN	22.1101	0.0784	0.9341	2.4760	0.9294	4.6319
SURE	37.5362	0.0133	0.9673	3.0975	0.9872	1.3319
PSRT	38.4961	0.0119	0.9752	2.5408	0.9919	1.0657
M2U-Net	40.8337	0.0091	0.9851	1.9447	0.9936	0.9627
CS2DIP	36.1736	0.0155	0.9706	3.0513	0.9838	1.5467
S^2CycleDiff	31.8571	0.0255	0.9711	2.8322	0.9711	1.7734
MTDiff	43.2506	0.0069	0.9897	1.5589	0.9948	0.8715

ablation is imposed on the benchmark dataset, Houston, for brevity and without affecting the generality.

1) Influence of the Dual-Resolution Diffusion Paradigm: In the diffusion learning stage, to sufficiently utilize the prior knowledge inherent in the input pairs LrHSI and HrMSI, the estimated degradation model is embedded into the diffusion model, which constructs the dual-resolution diffusion paradigm. With the supervised constraint under degraded resolution and the unsupervised regularization from the degradation model, better guidance is formed for diffusion learning. To evaluate the effectiveness of the dual-resolution diffusion paradigm, we construct a variant by removing the degraded-resolution branch. As illustrated in Table V, the dual-resolution diffusion paradigm obtains a better reconstruction of HrHSI than the native-resolution diffusion, which assesses the effectiveness of the dual-resolution diffusion paradigm.

2) Influence of the DCI Module: To validate our proposed DCI module, we conducted an extensive ablation study, with results detailed in Table V. We compared DCI against a carefully selected set of alternatives: a variant using the fast Fourier transform (FFT) to isolate the effect of the transform basis, the Fourier-enhanced (FE) module from FeINFN [6] as a powerful external benchmark, and a standard fully convolutional network (FCN) to establish a nonfrequency-based baseline.

The results show that, when operating under the proposed dual-resolution paradigm, the DCI module achieves the best performance across all six metrics, establishing the overall best results for this ablation study.

Furthermore, the comparison within the dual-resolution setting provides deeper insights. Notably, the FFT-based variant of our module consistently secures the second-best performance, significantly outperforming both the FE module and the standard FCN. This strongly suggests that the architectural design of our injection module is inherently more effective and robust for this task than other approaches. The superior performance of DCI over FFT can be attributed to the favorable properties of the discrete cosine transform, such as its energy compaction and avoidance of boundary artifacts, which make it the optimal choice for our framework.

In the simplified native-resolution-only setting, the results show a highly competitive landscape. While the FE module performs strongly, our DCI module remains highly competitive, securing the second-best results in most metrics and achieving the best ERGAS score. This again highlights that even under challenging conditions, our DCI module maintains robust, well-balanced performance, and it scales most effectively to achieve unparalleled results when empowered by our full-dual-resolution guidance system.

D. Parameters Analysis

In this section, to sufficiently evaluate our proposed MTDiff, a model parameter discussion is conducted to assess how hyperparameters affect MTDiff. Then, we show the analysis of complexity among all DL methods. Last, the robustness analysis is conducted which indicates the outstanding performance of the proposed MTDiff among various scale factors.

1) Model Parameters Discussion: The hyperparameters of diffusion models consist of the number of diffusion steps, the dropout rate, the depth of UDB, and the number of channels in the first UDB. However, for universality, the number of channels in the first UDB is based on the channels of LrHSI. Thus, the number of diffusion steps, dropout rate, and depth of UDB are experimented on the benchmark dataset, Houston, to evaluate the influence of hyperparameters.

First, the number of diffusion steps is assessed due to our proposed MTDiff taking the diffusion model as the backbone. MTDiff reconstructs HrHSI step-by-step along the diffusion track. To analyze the effect of the number of diffusion steps, we gradually shift it from 800 to 2000 and evaluate the performance in both spatial metrics, PSNR, and spectral metrics, SAM. As illustrated in Fig. 7(a), the performance of MTDiff remains stable while the steps change. As the number of diffusion steps increases, the noise-perturbed image gradually approximates a Gaussian distribution. Thus, the spatial performance becomes better. Furthermore, the dropout rate of the denoising network affects the information that is utilized from local features. To analyze the effect of the dropout rate, we gradually change it from 0.0 to 0.8. As shown in Fig. 7(b), the reconstruction quality becomes worse while the dropout rate increases, indicating that all information is important in the fusion task. Finally, the depth of UDB directly affects the complexity of the network, playing an important role. To evaluate the effect of the depth of UDB, we gradually add it from 1 to 3. As illustrated in Fig. 7(c), the quality of the fusion result becomes fidelity as the depth increases with a positive correlation. For the balance of the performance and cost of time, we set the number of diffusion steps to 1600, the number of UDB blocks to three, and the dropout rate to 0, in the following experiments among various datasets.

2) Model Complexity Analysis: An analysis of model complexity was conducted on the Houston dataset (see Table VI), through which a clear tradeoff between computational cost and performance is revealed. A higher parameter count and a longer inference time are observed for our proposed MTDiff when compared with one-shot methods. Notably, a significant speedup in inference is demonstrated by MTDiff when compared with S^2CycleDiff, another diffusion-based method, highlighting a more efficient design within its class. It is

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TABLE V

ABLATION STUDY WITH SIX QUANTITATIVE METRICS ON THE HOUSTON DATASET. THIS STUDY EVALUATES TWO KEY COMPONENTS OF MTDIFF: THE DIFFUSION PARADIGM AND THE HIGH-FREQUENCY OPTIMIZATION MODULE. TO VALIDATE OUR PROPOSED DCI MODULE, WE COMPARE IT AGAINST THREE ALTERNATIVES: A STANDARD FCN, A MODULE BASED ON THE FFT, AND AN FE STRATEGY [6]. THE BEST AND SECOND-BEST RESULTS WITHIN EACH PARADIGM ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY. THE OVERALL BEST PERFORMANCE IS MARKED WITH BOLD

Diffusion Paradigm	Injection Module	PSNR	RMSE	SSIM	SAM	UQI	ERGAS
BEST		∞	0	1	0	1	0
Native-resolution (w/o Degraded Branch)	FCN	30.2113	0.0309	0.8620	8.2151	0.9299	4.3365
	FFT	29.7999	0.0324	0.8503	8.7158	0.9268	5.1555
	FE	31.2878	0.0273	0.8844	7.3155	0.9404	4.8677
	DCI	31.0766	0.0279	0.8777	7.5123	0.9374	4.3293
Dual-resolution	FCN	42.9071	0.0072	0.9914	2.0070	0.9886	0.9979
	FFT	43.5850	0.0066	0.9926	1.8488	0.9897	0.8829
	FE	43.1842	0.0069	0.9914	1.9708	0.9888	0.9738
	DCI	**43.6221**	**0.0066**	**0.9928**	**1.8293**	**0.9898**	**0.8532**

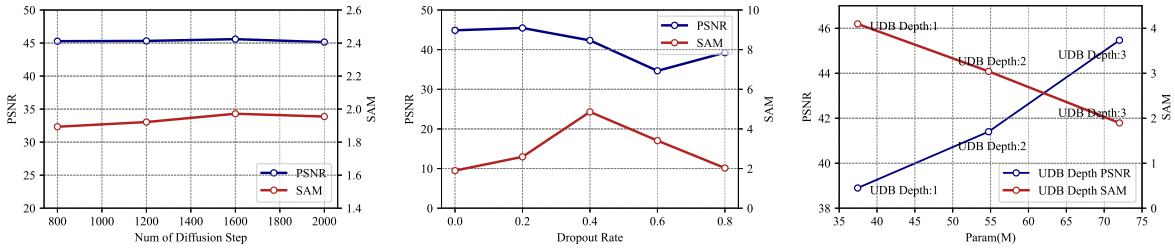


Fig. 7. Parameters discussion on the Houston dataset. Each subfigure gives the change of PSNR and SAM metrics based on the shifting for different hyperparameters. (a) Parameter analysis of diffusion steps. (b) Parameter analysis of dropout rate. (c) Parameter analysis of depth of UDB.

TABLE VI

COMPLEXITY AND INFERENCE TIME ANALYSIS OF DL METHODS DURING INFERENCE UPON HOUSTON DATASET

Method	Parameter Size(M)	Time per Iteration(s)	Inference Time(s)
UDALN	0.02	0.006	0.003
SURE	0.18	0.080	0.009
PSRT	0.55	0.216	0.059
M2U-Net	14.70	0.405	0.125
CS2DIP	8.54	0.083	0.027
S^2CycleDiff	36.27	0.124	24.758
MTDiff	72.04	0.041	4.640

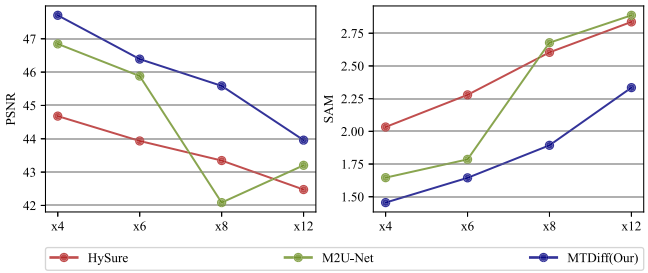


Fig. 8. PSNR and SAM metrics under various scale factors from 4 to 12 on the benchmark dataset Houston.

posited that this tradeoff is justified for accuracy-critical applications where performance is paramount. This is considered an inherent consequence of its iterative diffusion mechanism, where the target image is reconstructed step-by-step. Through this process, the target data distribution is meticulously modeled, leading to the superior fidelity and robustness demonstrated in our results. The increased complexity is therefore presented as a direct tradeoff for achieving state-of-the-art performance. While this tradeoff is justified for accuracy-critical applications, it concurrently implies that MTDiff may not be the optimal choice for resource-constrained or real-time scenarios.

3) Robustness Analysis: To evaluate the robustness of our proposed MTDiff, we examine the influence of various scale factors on the benchmark dataset, Houston, for brevity and without affecting the generality. From conventional to DL-based, two representative methods (HySure and M2U-Net) are compared with our proposed MTDiff. The spatial metrics PSNR and spectral metrics SAM of the fusion results under four different scale factors are illustrated in Fig. 8. The variance of the scale factor leads to different influences on each method. The DL method M2U-Net is more sensitive to the scale factor, with different performance in various scale factors. On the contrary, the matrix factorization method, HySure, is less sensitive while facing changes in the scale factor. Furthermore, both PSNR and SAM metrics of our proposed MTDiff keep the best value among all comparison methods and intuitively present linear variance based on changing the scale factor, which demonstrates the excellent robustness of MTDiff.

E. Validation of Degradation Model Estimation

To address potential concerns about the overfitting and physical plausibility of our degradation model estimation (as

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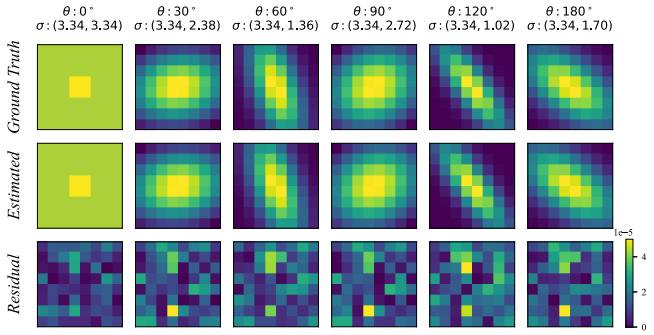


Fig. 9. Qualitative comparison of estimated spatial degradation kernels. The first row shows the GT 2-D Gaussian kernels with varying rotation angles (θ) and standard deviations (σ). The second row displays the corresponding kernels estimated by our SpaNet. The third row visualizes the pixelwise residual error.

discussed in Section III-B), we conducted a series of validation experiments. These experiments are designed to demonstrate that our method can accurately learn the underlying physical degradation processes and generalize well to unseen data, thus providing a reliable foundation for the subsequent fusion task.

1) Qualitative Analysis of PSF Estimation: To qualitatively assess the ability of our SpaNet to estimate the spatial PSF, we generated several synthetic datasets where the GT PSF is a 2-D Gaussian kernel with varying degrees of anisotropy and rotation angles (θ). For each benchmark dataset, we partitioned the data into a training set (57%) and a hold-out testing set (43%). As illustrated in Fig. 9, our estimated kernels (second row) closely match the GT kernels (first row) across all variations. The residual maps (third row) show minimal error, with magnitudes on the order of 10^{-5}, confirming that our physically constrained SpaNet can accurately capture a wide range of realistic blurring effects without prior knowledge of the specific kernel parameters. This visual evidence supports the claim that our approach learns a physically meaningful transformation rather than an arbitrary function.

2) Quantitative Analysis of Generalization: To quantitatively evaluate the generalization capability of our degradation model, we performed experiments on a split-dataset setup. For each benchmark dataset, we partitioned the data into a training set (57%) and a hold-out testing set (43%). The SpaNet and SpeNet were trained exclusively on the training set. Subsequently, these trained operators were applied to the unseen test set to generate the degraded images, which were then compared against the GT degradations.

Table VII presents the evaluation metrics for this generalization task. The results are exceptional across all datasets. For instance, when generating the HrMSI from HrHSI shown in Table VII(a), the PSNR values are consistently above 69 dB. Similarly, when generating the LrHSI from HrHSI illustrated in Table VII(b), the PSNR values are even higher, exceeding 88 dB. The near-perfect SSIM scores (1.0000) and extremely low RMSE, SAM, and ERGAS values further corroborate that the degradation model, trained on a separate part of the data, generalizes almost perfectly to the testing set.

TABLE VII

QUANTITATIVE EVALUATION OF THE DEGRADATION MODEL'S GENERALIZATION ON HOLD-OUT TESTING SETS. THE MODELS WERE TRAINED ON 53% OF THE DATA AND EVALUATED ON THE REMAINING 47%. (A) PERFORMANCE OF GENERATING HRMSI FROM HRHSI. (B) PERFORMANCE OF GENERATING LRHSI FROM HRHSI

Dataset	PSNR	RMSE	SSIM	SAM	UQI	ERGAS
BEST	∞	0	1	0	1	0
chikusei	71.1344	0.0003	1.0000	0.0460	0.9998	0.0515
houston	71.1493	0.0003	1.0000	0.0591	0.9999	0.0213
washington	69.3476	0.0003	1.0000	0.0669	0.9999	0.0166

(a)

Dataset	PSNR	RMSE	SSIM	SAM	UQI	ERGAS
BEST	∞	0	1	0	1	0
chikusei	88.3108	0.0000	1.0000	0.0060	0.9933	0.0043
houston	102.2394	0.0000	1.0000	0.0059	0.9851	0.0006
washington	99.6056	0.0000	1.0000	0.0059	0.9958	0.0007

(b)

In conclusion, both the qualitative visualization and the quantitative generalization experiments provide strong evidence that our constrained degradation model estimation method is robust and reliable. It effectively avoids overfitting and learns physically plausible operators, thereby validating its suitability for our unsupervised fusion framework.

V. CONCLUSION

In this article, a model-embedded two-stage diffusion method (MTDiff) is proposed for unsupervised multispectral and HSI fusion. Stage one introduces a degradation model estimation to extract the prior information from input pairs. Stage two unsupervisedly reconstructs the HrHSI via the dual-resolution diffusion model embedded with an estimated degradation model. Specifically, treating the degradation process as a fixed diffusion step, the unsupervised diffusion paradigm is innovatively realized based on the mapping from upsampled LrHSI to HrHSI. Moreover, a dual-resolution diffusion model is designed for simultaneous noise-perturbed diffusion across native and degraded resolution, achieving per-step guidance under physical constraints and prior knowledge. In addition, a DCI module is introduced to flatten the frequency information to clear the 2-D domain with a huge high-frequency area, constructing a high-frequency shortcut for learning, which achieves sharp textures and clear structures in the fusion result. Extensive experiments among three benchmark datasets indicate our proposed MTDiff's favorable performance against the other nine fusion methods, assessing its superiority quantitatively and qualitatively. Notwithstanding its superior performance, the proposed MTDiff framework has limitations that open up avenues for future exploration. The most prominent limitation is the computational burden inherent in diffusion models. The necessity of a large number of iterative steps for high-fidelity reconstruction leads to substantial inference time, which may constrain its application in time-sensitive scenarios.

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